

# A 14-bit 70MS/s pipeline ADC with power-efficient back-end stages

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**Abstract**—A 14-bit SHA-less pipeline ADC based on 2.5-bit per stage architecture is proposed. Implemented in TSMC 0.18 $\mu$ m CMOS process, the proposed ADC achieves an ENOB of 11.34-bits and consumes 41 mW power at 70 MS/s thereby achieving an FOM of 226fJ/conversion-step which compares favorably with state-of-the-art work. The low-power consumption is attributed to the careful design of back-end stages based on a novel gain-boosting recycling folded-cascode amplifier.

**Keywords** - SHA-less pipeline ADC, Recycling folded cascode amplifier, Gain boosting.

## I. INTRODUCTION

Pipeline ADCs are the architecture of choice for implementing wireless communication systems because of their low-power operation to simultaneously achieve high resolution (12-16 bits) and high-performance (70-80 dB SNDR) at high-speed (100-300 MS/s) analog to digital conversion [1]. Low-power ADC realization is highly desirable for all applications especially for battery operated portable systems. The aforementioned problem has motivated many researchers to decrease the power consumption while implementing robust and high performance pipeline ADCs. [2] and [3] proposed multi-bit per stage pipeline architecture to reduce power while [4] implemented opamp sharing among two stages to further decrease power consumption. [5] used digitally assisted analog approach by implementing low-power open-loop amplification and using digital background calibration to continuously estimate and remove the resultant open-loop gain errors.

Since operational transconductance amplifier (OTA) which forms an integral part of MDAC consumes the most of ADC power, all efforts targeting low-power operation aims to save power in this critical block. Single-stage amplifiers are usually the preferred design choice over their two-stage counterparts for low-power realization. Among single-stage amplifiers, telescopic and folded-cascode are the two most widely used OTA topologies for implementing high-speed and high-resolution pipeline ADC. Although telescopic OTA consumes less power than folded-cascode, it suffers from restricted output swing making it unfavorable for modern low-voltage CMOS processes. To improve the power efficiency of folded-cascode amplifier, [6] proposed Recycling Folded Cascode (RFC) amplifier which recycles the tail current back into the input differential pair and boost the input transconductance to double its unity-gain frequency with moderate improvements in DC gain, slew-rate and input

referred noise.

Our ADC is based on the aforementioned recycling folded-cascode amplifier with added gain-boosting to satisfy the required settling-time requirements with minimum current consumption. The basic motivation of this work is to meet the target specifications (70dB SNDR at 70MS/s sampling speed) with minimum power using calibration-free architecture. The paper is organized as follows: In section II, the proposed ADC architecture is explained along with novel implementation of power-efficient back-end stages. Circuit-level implementation is discussed in section III while section IV covers the simulation results and comparison summary of our paper with recent work. Section V concludes the paper.

## II. PROPOSED ADC ARCHITECTURE

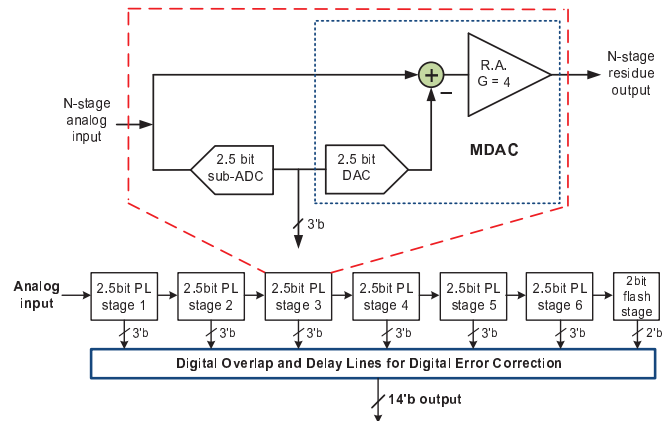


Fig. 1. Architectural diagram of 14bits SHA-less Pipeline ADC implemented as six 2.5-bit pipeline stages and single 2-bit flash stage.

Targeting low-power consumption, we design a 14-bit SHA-less pipeline ADC using six fully-differential 2.5-bit stages and one 2-bit flash ADC as shown in Fig. 1. The first two stages are based on traditional gain-boosting folded-cascode amplifiers where the gain-booster auxiliary amplifiers are also implemented by folded-cascode design. However the last four 2.5-bit stages have been implemented using novel gain-boosting structure which consumes less power than conventional gain-boosting architecture. Fig. 2 shows schematic of the residue amplifier used to implement 2.5-bit

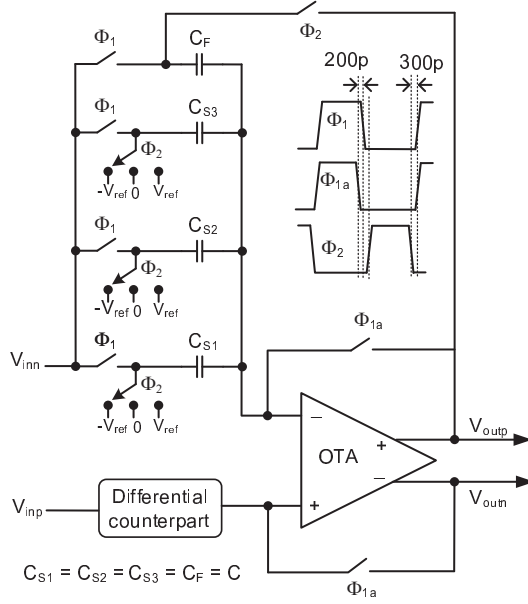


Fig. 2. Schematic of the residue amplifier for 2.5-bit MDAC. Shown single-ended but implemented as fully-differential. Also shown are the three clock phases with non-overlapping time as 300ps and advanced clock edge by 200ps.

Unlike conventional gain-boosting amplifiers, which employ two fully-differential auxiliary amplifiers to boost the output resistance of the cascodes in main amplifier as shown in Fig. 4, the proposed architecture makes use of four simple common-source amplifiers to achieve gain-boosting by increasing the resistance of individual cascodes of the main amplifier. The novelty in backend ADC is based on two innovations. First, a recycling folded-cascode amplifier is used to achieve better power efficiency than conventional folded-cascode amplifier. Secondly, two fully-differential auxiliary amplifiers (total transistor count being 22) are replaced by four low-power and simplified CS-amplifiers (with total transistor count only 8).

Each 2.5-bit pipeline stage output 3-bit digital code, which when combined with 2-bits of the last flash-ADC stage makes the output 20-bit long. The raw bits are then processed via RSD algorithm [2] in digital error correction (DEC) block to evaluate 14-bit final ADC output. The DEC block is also implemented on-chip. Each of the 2.5-bit sub-ADC resolve 3 digital bits via six comparators. In addition to be sent to the DEC block, these binary bits are also used to implement DAC switching scheme to choose appropriate reference voltage and thus calculate the residue voltage for next stage.

As a fair trade-off between complexity and low-power implementation, first stage has been designed for 14-bit accuracy, second stage with 12-bit accuracy and last four stages have been designed identically with 10-bit accuracy each. In the last four MDAC, the residue amplifier is realized with a novel gain-boosting recycling folded cascode amplifier discussed in next section.

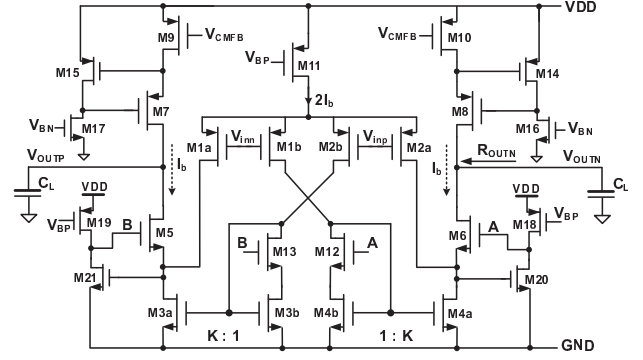


Fig. 3. Schematic of the novel gain-boosted recycling folded-cascode amplifier used in the residue amplifier within MDAC implementation of back-end stages 3, 4, 5 and 6

The pipeline ADC was implemented in TSMC 0.18 $\mu$ m CMOS process. The proposed gain-boosting recycling folded-cascode amplifier which forms the basis of residue amplifier in multiplying DAC of stage 3,4,5 and 6 is shown in Fig. 3. Compared to conventional folded-cascode amplifier, the input differential pair is split in two equal halves  $M1a$ ,  $M1b$  and  $M2a$ ,  $M2b$  while the tail current-source transistors  $M3$  and  $M4$  are split by  $K : 1$  to form two current mirrors  $M3a - M3b$  and  $M4a - M4b$ . For optimal trade-off between power-efficiency and phase-margin, we choose  $K = 3$  for our design.

In switched-capacitor circuits, settling speed is determined by the unity-gain frequency and phase-margin while settling accuracy depends upon DC-gain of the OTA. To achieve the required settling accuracy and obtain precise residue multiplication (4X) in back-end ADC, gain-boosting is essential which is thus implemented using common-source amplifiers  $M14$ ,  $M15$ ,  $M20$ ,  $M21$  biased by low-threshold transistors  $M16$ ,  $M17$ ,  $M18$  and  $M19$  respectively. These single-transistor common-source amplifiers boost the resistance and hence gain of the cascode pairs  $M5$ ,  $M6$ ,  $M7$ ,  $M8$  using regulated-cascode principle [7]. Compared to fully-differential auxiliary amplifier, this approach not only simplifies the design but also reduces power. The total output resistance  $R_{OUTN}$  as seen from the output  $V_{outn}$  is given by:

$$R_{out} = (g_{m6}r_{ds6}(1 + A_{M20}) + 1)(r_{ds4a}||r_{ds2a})|| \frac{(g_{m8}r_{ds8}(1 + A_{M14}) + 1)r_{ds10}}{(g_{m8}r_{ds8}(1 + A_{M14}) + 1)r_{ds10}} \quad (1)$$

where  $A_{M14}$  and  $A_{M20}$  are the gain of auxiliary common-source amplifiers  $M14$  and  $M20$  respectively.

The schematic of the switched-capacitor comparator is adopted from [8] and consists of switched capacitors followed by a cross-coupled latch as shown in Fig. 5. The latch is implemented by a double-tail comparator for its higher speed and lower kick-back noise [9]. At phase  $\Phi_1$ , capacitor  $C_{s1}$  samples the reference voltages. Bottom plate sampling is enabled by turning off the bottom reference voltage slightly earlier through phase  $\Phi_{1a}$ . At phase  $\Phi_2$ , the switched-capacitors subtract

reference voltage from the input and comparison is made at the rising edge of  $\Phi_{2a,b,ar}$  (or falling edge of  $\Phi_{2a}$ ). The differential voltage presented to the latch inputs is given by equation (2). The latch is realized with double-tail comparator along-with SR latch to keep the output valid for one complete cycle.

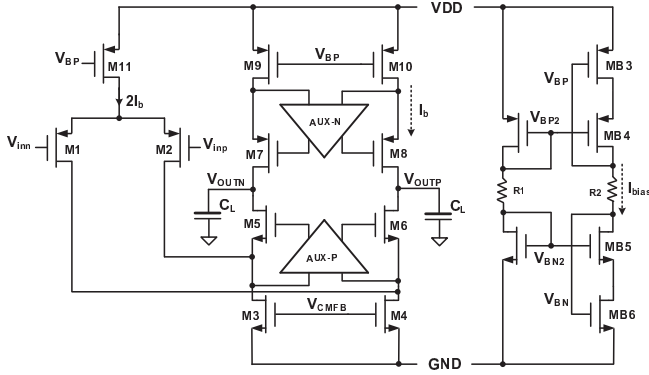


Fig. 4. Schematic of the conventional gain-boostered folded-cascode amplifier used in the residue amplifier of MDAC in the first two pipeline stages. High-swing cascode bias circuit is shown on the right.

$$V_{diff} = (V_{inp} - V_{inn}) - \frac{C_{s1}}{C_{s1} + C_{s2}} \cdot (V_{REFP} - V_{REFN}) \quad (2)$$

Each sub-ADC features six comparators to resolve the six comparison thresholds i.e.  $\pm\frac{1}{8}$ ,  $\pm\frac{1}{4}$  and  $\pm\frac{3}{8}$ . Due to RSD algorithmic redundancy, each comparator can tolerate maximum offset of  $\pm 125mV$ . Since, we do not employ front-end sample and hold circuit to save power, the comparators were designed for  $3\sigma$  offset of around  $\pm 50mV$  and the remaining tolerance is used to absorb the sampling mismatch between MDAC and sub-ADC of the first stage. Capacitor ratio is adjusted to realize different switching thresholds. For instance, to resolve  $+\frac{1}{8}$ ,  $\frac{C_{s1}}{C_{s2}}$  is chosen to be  $\frac{1}{7}$  or more precisely  $\frac{20f}{140f}$ . Similarly, to resolve  $-\frac{1}{8}$ , the position of  $V_{REFP}$  and  $V_{REFN}$  is swapped.

The 6-bit comparators output is a thermometer code which is converted into 3-bit binary code through thermometer-to-binary converter and bubble-error removal logic. The 3-bit binary output from each sub-ADC or pipeline stage serves dual purpose. First, this code is used to control DAC switches and select appropriate reference voltage for the bottom plate sampling capacitors of the residue amplifier in each MDAC stage. Second, these binary bits are taken to digital-error-correction block where they are combined with the binary output of other pipeline stages and added after necessary time-alignment to generate the final 14-bit ADC output.

The ADC timing is shown in Fig. 6. First MDAC samples the input on the falling edge of  $\Phi_2$ . At the same instant, the six comparators of first sub-ADC also sample the input and subtract from the reference that had been previously sampled on the falling edge of  $\Phi_1$ . During next half cycle of  $\Phi_1$  the MDAC configures in integration mode where it implements the

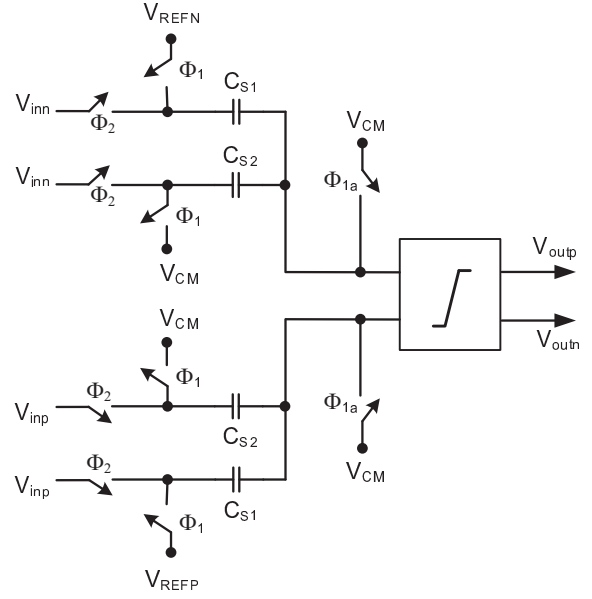


Fig. 5. Schematic of the switched-capacitor comparator.

function  $4 \times Vin \pm d \times V_{ref}$  where  $d = 0, \pm 1, \pm 2, \pm 3$  depending upon 3-bit binary output from the sub-ADC. By the end of  $\Phi_1$ , the integration phase for first MDAC is completed and second MDAC along-with its sub-ADC comparators sample the residue voltage output by the first stage at the falling edge of  $\Phi_1$ . Similarly all other pipeline stages sample on alternate falling edges of  $\Phi_1$  and  $\Phi_2$  so that by the end of forth  $\Phi_2$  cycle all stages had sampled their first input as shown in the timing diagram. It is important to note that only first stage MDAC and sub-ADC capacitors see time-varying input while the next stages all see a constant residue voltage for the duration of their sampling phase. For this reason, the sampling paths of the first MDAC and sub-ADC capacitors should be carefully matched to avoid any difference in the sampled voltage which may cause relative aperture error and thus distortion at higher input frequencies.

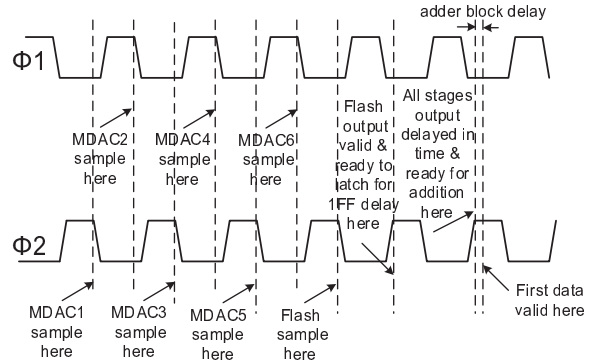


Fig. 6. Schematic of the switched-capacitor comparator. The capacitor ratios and reference voltage polarity were adjusted to generate six comparator-threshold-voltages.

#### IV. SIMULATION RESULTS

Realized in TSMC  $0.18\mu m$  CMOS process, the prototype ADC was carefully laid-out taking into account precision mixed-signal layout guidelines. Techniques such as common-centroid layout, dummy insertion and guard rings are readily employed to minimize offset and other non-idealities. Post-layout simulations were carried out across process corners to ascertain the post-silicon robust performance of implemented ADC. As shown in Fig. 7 our ADC achieves 11.34 bits of ENOB while sampling at  $70MS/s$  and consuming 41mW of power with corresponding FOM of 226fJ/conv-step. The comparison of our ADC is made with published work as shown in table I. For fair comparison, we only consider the pipeline ADCs implemented in  $0.18\mu m$  CMOS process and sampling rates of 50MHz or above. The FOM used for the comparison is Walden or ISSCC FOM which relates the ADC power consumption to its performance; measured by sampling-rate and conversion-error amplitude and defined as:

$$FOM = \frac{P}{2^{ENOB} \cdot f_s} \quad (3)$$

where  $P$  is the power consumption,  $f_s$  is the sampling frequency and  $ENOB$  is the effective number of bits defined by the signal-to-noise and-distortion-ratio (SNDR) as:

$$ENOB = \frac{SNDR - 1.76}{6.02} \quad (4)$$

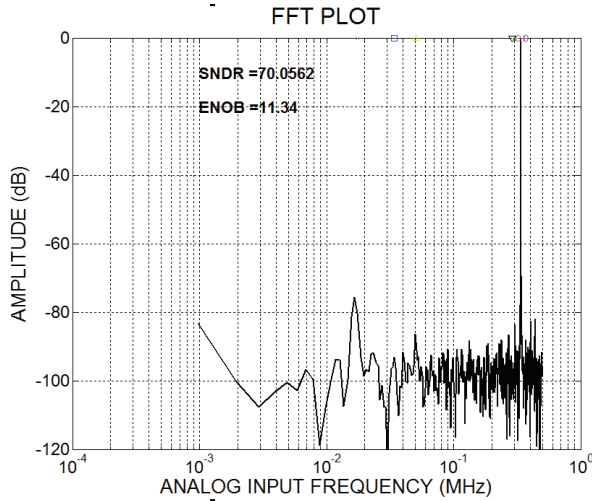


Fig. 7. 1024 point FFT with  $-1dBFS$  input at  $f_{in} = 23MHz$ ,  $f_s = 70MS/s$ .

TABLE I. PERFORMANCE COMPARISON OF OUR ADC WITH PREVIOUS WORK.

| Reference | Resolution [bits] | Speed [MS/s] | ENOB [bits] | Power [mW] | FOM [fJ/step] |
|-----------|-------------------|--------------|-------------|------------|---------------|
| [10]      | 16                | 125          | 12.5        | 385        | 532           |
| [11]      | 14                | 100          | 11.73       | 230        | 677           |
| [12]      | 10                | 50           | 9.4         | 9.9        | 300           |
| [13]      | 10                | 50           | 8.78        | 18         | 819           |
| [14]      | 8                 | 200          | 6.4         | 8.5        | 503           |
| This work | 14                | 70           | 11.34       | 41         | 226           |

#### V. CONCLUSION

We present a 14-bit SHA-less pipeline ADC that features four very low-power 2.5-bit back-end stages. The key idea is the use of a novel gain-boosting recycling folded-cascode amplifier in back-end stages. Implemented in  $0.18\mu m$  CMOS process, our ADC achieves 70dB SNDR while sampling at  $70MS/s$  and consuming around 41mW power with a competitive FOM of 226fJ/conv-step.

#### ACKNOWLEDGMENT

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#### REFERENCES

- [1] Walden, R.H., "Analog-to-digital converter survey and analysis," Selected Areas in Communications, IEEE Journal on , vol.17, no.4, pp.539,550, Apr 1999.
- [2] Lewis, S.H.; Gray, P.R., "A pipelined 5-Msample/s 9-bit analog-to-digital converter," Solid-State Circuits, IEEE Journal of , vol.22, no.6, pp.954,961, Dec 1987.
- [3] Singer, L.A.; Brooks, T.L., "A 14-bit 10-MHz calibration-free CMOS pipelined A/D converter," VLSI Circuits, 1996. Digest of Technical Papers., 1996 Symposium on , vol., no., pp.94,95, 13-15 June 1996.
- [4] Nagaraj, K.; Fetterman, H.S.; Anidjar, J.; Lewis, S.H.; Renninger, R.G., "A 250-mW, 8-b, 52-Msamples/s parallel-pipelined A/D converter with reduced number of amplifiers," Solid-State Circuits, IEEE Journal of , vol.32, no.3, pp.312,320, Mar 1997.
- [5] Murmann, B.; Boser, B.E., "A 12-bit 75-MS/s pipelined ADC using open-loop residue amplification," Solid-State Circuits, IEEE Journal of , vol.38, no.12, pp.2040,2050, Dec. 2003.
- [6] Assaad, R.S.; Silva-Martinez, J., "The Recycling Folded Cascode: A General Enhancement of the Folded Cascode Amplifier," Solid-State Circuits, IEEE Journal of , vol.44, no.9, pp.2535,2542, Sept. 2009.
- [7] Bult, K.; Geelen, G.J.G.M., "A fast-settling CMOS op amp for SC circuits with 90-dB DC gain," Solid-State Circuits, IEEE Journal of , vol.25, no.6, pp.1379,1384, Dec 1990.
- [8] Kehan Zhu; Balagopal, S.; Saxena, V., "Systematic design of 10-bit 50MS/s pipelined ADC," Microelectronics and Electron Devices (WMED), 2013 IEEE Workshop on , vol., no., pp.17,20, 12-12 April 2013.
- [9] Schinkel, D.; Mensink, E.; Klumperink, E.; van Tuijl, E.; Nauta, B., "A Double-Tail Latch-Type Voltage Sense Amplifier with 18ps Setup+Hold Time," Solid-State Circuits Conference, 2007. ISSCC 2007. Digest of Technical Papers. IEEE International , vol., no., pp.314,605, 11-15 Feb. 2007.
- [10] Devarajan, S.; Singer, L.; Kelly, D.; Decker, S.; Kamath, A.; Wilkins, P., "A 16-bit, 125 MS/s, 385 mW, 78.7 dB SNR CMOS Pipeline ADC," Solid-State Circuits, IEEE Journal of , vol.44, no.12, pp.3305,3313, Dec. 2009.
- [11] Byung-Geun Lee; Byung-Moo Min; Manganaro, G.; Valvano, J.W., "A 14b 100MS/s Pipelined ADC with a Merged Active S/H and First MDAC," Solid-State Circuits Conference, 2008. ISSCC 2008. Digest of Technical Papers. IEEE International , vol., no., pp.248,611, 3-7 Feb. 2008.
- [12] Ahmed, I.; Mulder, J.; Johns, D.A., "A Low-Power Capacitive Charge Pump Based Pipelined ADC," Solid-State Circuits, IEEE Journal of , vol.45, no.5, pp.1016,1027, May 2010.
- [13] Seung-Tak Ryu; Bang-Sup Song; Bacrania, K., "A 10b 50MS/s pipelined ADC with opamp current reuse," Solid-State Circuits Conference, 2006. ISSCC 2006. Digest of Technical Papers. IEEE International , vol., no., pp.792,801, 6-9 Feb. 2006.
- [14] Brooks, L.; Hae-Seung Lee, "A Zero-Crossing-Based 8b 200MS/s Pipelined ADC," Solid-State Circuits Conference, 2007. ISSCC 2007. Digest of Technical Papers. IEEE International , vol., no., pp.460,615, 11-15 Feb. 2007